

WEEK 1: MEET PYTHON & SETUP

Day 1: Welcome to Coding & Setup

What a program is, and the Input → Process → Output idea. Installing Python with the Thonny editor and touring the editor and the Shell. Writing a first program: `print("Hello, KidCode!")`.

Day 2: Print, Escapes & Comments

Mastering `print()`, fixing quote errors, and using escape characters (`\n`, `\t`, `\'`). Adding `#` comments. Lab: "Name Banner" ASCII art.

Day 3: Variables: Labeled Boxes

Storing information with `=`. Variable naming rules and `snake_case`, changing values, and checking data types with `type()` (`str`, `int`, `float`). Lab: "About Me Card" built from variables.

Day 4: Number Crunching

Arithmetic operators (`+`, `-`, `*`, `/`, `//`, `%`, `**`) and PEMDAS. Integers vs. floats, and the `TypeError` from mixing text and numbers. Lab: "Snack Shop Calculator".

Day 5: Talking to the User

Asking questions with `input()`, converting answers with `int()` and `float()`, and formatting output with f-strings like `f"Hi, {name}!"`. Practice: "Age in Dog Years" and "Tip Helper".

WEEK 2: WORDS AS DATA

Day 6: Lab: Mad Libs Machine

Project 1. Combining `input()`, variables, and f-strings into a complete program: a silly story with blanks that players fill in. Pair programming with driver and navigator roles.

Day 7: Strings: Beads on a String

Strings as ordered characters. Measuring with `len()`, joining with `+`, repeating with `*`, and picking characters with `word[0]` and `word[-1]`. Understanding `IndexError`. Practice: "Secret Initials".

Day 8: Slicing & Dicing Text

Slicing with `[start:end]` and reversing with `[::-1]`. Why strings are immutable, so new strings are built instead of edited. Practice: "Nickname Maker" and "Backwards Words".

Day 9: String Methods: Dot Magic

The idea of a method: an action that belongs to a piece of data, called with a dot. Using `.upper()`, `.lower()`, `.title()`, `.strip()`, `.replace()`, and `.count()`. Lab: "Text Cleaner".

Day 10: More Methods & Text Toolbox

Using `.split()`, `.join()`, `.find()`, `.startswith()`, and `in`. Lab: "Text Toolbox", which reports a sentence's length, word count, and shout and whisper versions.

WEEK 3: MAKING DECISIONS

Day 11: Booleans & Comparisons

The `bool` type, comparison operators (`==`, `!=`, `<`, `>`, `<=`, `>=`), and the difference between `=` and `==`. "Predict, Run, Explain" cards.

Day 12: if & else: Choosing a Path

Writing `if` and `else` blocks, colons, and indentation as Python's grammar. Fixing `IndentationError`. Practice: "Password Checker" and "Can You Ride?", a theme park height checker.

Day 13: elif, and, or, not

Multi-way choices with `elif`, combining conditions with `and`, `or`, and `not`, and placing an `if` inside another `if`. Practice: "Grade Calculator" and "Movie Night Picker".

Day 14: Bug Hunters: Debugging Skills

Reading error messages (`SyntaxError`, `NameError`, `TypeError`) from the bottom line up. Print debugging and stepping through code in Thonny. Lab: "Bug Zoo", ten broken programs to fix.

Day 15: Lab: Mystic Oracle

Project 2. A first library: `import random`, `random.randint()`, and `random.choice()`. A fortune-telling program with `if/elif` logic. Big idea: `import` borrows code someone else wrote.

WEEK 4: LOOPS

Day 16: while Loops: Repeat Until

Repeating code with `while`, loop conditions, and counters. Stopping runaway loops with Thonny's Stop button. Practice: "Countdown Rocket" and "Three Tries Password".

Day 17: for Loops & range()

Looping with `for` and `range()` using one, two, and three arguments. Choosing between `for` (known count) and `while` (unknown count). Practice: "Times Table Printer".

Day 18: Counting & Looping Through Text

The accumulator pattern for running totals and counters, and looping through every character of a string. Practice: "Vowel Counter", "Palindrome Checker", and "Average Calculator".

Day 19: break, continue & Nested Loops

Leaving a loop early with `break`, skipping a round with `continue`, and loops inside loops for grids and patterns. Practice: "Multiplication Grid" and "Star Triangle".

Day 20: Lab: Guess the Number Deluxe

Project 3. Combining `while`, `if/elif/else`, `random.randint()`, and counters into a game with hints, a guess counter, and a play-again option.

WEEK 5: LISTS & TUPLES

Day 21: Lists: Organizing Many Things

Creating lists with `[]`, zero-based indexing, negative indexes, `len()`, and slicing with `[start:end]`. Comparing lists with strings. Practice: "Top 5 Ranker".

Day 22: List Methods & Mutability

Changing lists in place with `.append()`, `.insert()`, `.remove()`, `.pop()`, `.sort()`, and `.reverse()`. Membership tests with `in` and `not in`. Practice: "Playlist Editor".

Day 23: Looping Over Lists & Tuples

Using `for item in my_list`, building new lists in a loop, and finding totals, maximums, and minimums. Tuples as read-only lists, and unpacking like `x, y = point`.

Day 24: Words in a List

Turning text into lists with `.split()` and back into text with `.join()`. Sorting words and finding the longest word. Lab: "Sentence Scrambler" that shuffles a sentence.

Day 25: Lab: Wish List Manager

Project 4. A menu-driven program that combines `while`, `input()`, `if/elif`, and lists to add, remove, and show items. Design the menu on paper first, then test in pairs.

WEEK 6: DICTIONARIES & DATA

Day 26: Dictionaries: Key-Value Pairs

Creating dictionaries with `{}`, looking up values by key, and adding, changing, and deleting entries. Checking keys with `in`, and why `KeyError` appears. Practice: "Pet Profile".

Day 27: Looping Through Dictionaries

Using `.keys()`, `.values()`, `.items()`, and `.get()` with default values. Counting with dictionaries and choosing between a list and a dictionary.

Day 28: Lab: Word Counter

Using `.split()`, loops, and a dictionary to count how often each word appears in a sentence, then finding the most common word. A first taste of text analysis.

Day 29: Lists of Dictionaries

Storing many records, such as students or trivia questions, as a list of dictionaries. Looping through records and reading fields by key. Practice: "Class Directory".

Day 30: Lab: Trivia Quiz Show

Project 5. Questions and answers stored in a list of dictionaries, with score keeping. Accepting messy answers with `.lower()` and `.strip()`, and printing the score with an f-string.

WEEK 7: FUNCTIONS

Day 31: Functions: Reusable Recipes

Defining functions with `def`, calling them, and why functions keep code tidy (Don't Repeat Yourself). Reading function definitions like recipes. Practice: "Greeting Machine".

Day 32: Parameters & Arguments

Passing information into functions with one or more parameters and default values. Positional and keyword arguments. Practice: "Area Calculator" and "Custom Cheer".

Day 33: return Values

`return` versus `print()`, storing a function's result in a variable, and returning `True` or `False`. Practice: "Even or Odd?" and "Is It a Palindrome?" as functions.

Day 34: Scope, Docstrings & Testing

Local vs. global variables, docstrings that explain a function, and testing with sample inputs and expected outputs. Lab: "Function Detective" fixes broken functions.

Day 35: Lab: Refactor Day

Rewriting older programs, including "Word Counter" and "Trivia Quiz Show", as clean functions with clear names. Comparing before-and-after code with a partner.

WEEK 8: CAPSTONE & SHOWCASE

Day 36: Capstone I: Haunted School Plan

Project 6. Planning a text adventure called "Escape the Haunted School": sketching the map, listing rooms, items, and win and lose conditions, and drawing a flowchart of the game loop.

Day 37: Capstone II: Rooms & Movement

Storing rooms in a dictionary, moving with `go north` commands, and writing the `while` game loop. Handling unknown commands politely, with a helper function for each action.

Day 38: Capstone III: Items & Inventory

Adding an inventory list, `take` and `use` commands, locked doors, and a win condition. Testing each feature as it is added and keeping a bug list.

Day 39: Playtest & Polish

Peer playtesting with "Two Stars and a Wish" feedback cards, bug fixing, and adding room descriptions and ASCII art. Stretch goals: health points and a secret ending.

Day 40: Showcase Day & Year Review

Presenting games to classmates and guests. A "Python Trivia" review relay of Weeks 1 to 8, a skills checklist, and a preview of Year 2: files, classes, and libraries.